Session 7: Unsupervised Learning. Clustering

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Objectives

- Study the **basic concepts**, methods, and applications of cluster analysis.
- Review different **distance or similarity measures** for cluster analysis.
- Learn the most well-known **partitioning algorithms** based on distance, including K-means, K-medians, and kernel K-means algorithms.
- Learn *hierarchical clustering algorithms* (Aglomerative algorithm), and the Gaussian approach.
- Study the concepts and methods for **clustering evaluation and validation** using external, internal measures, and measures to assess the stability and tendency in clustering.

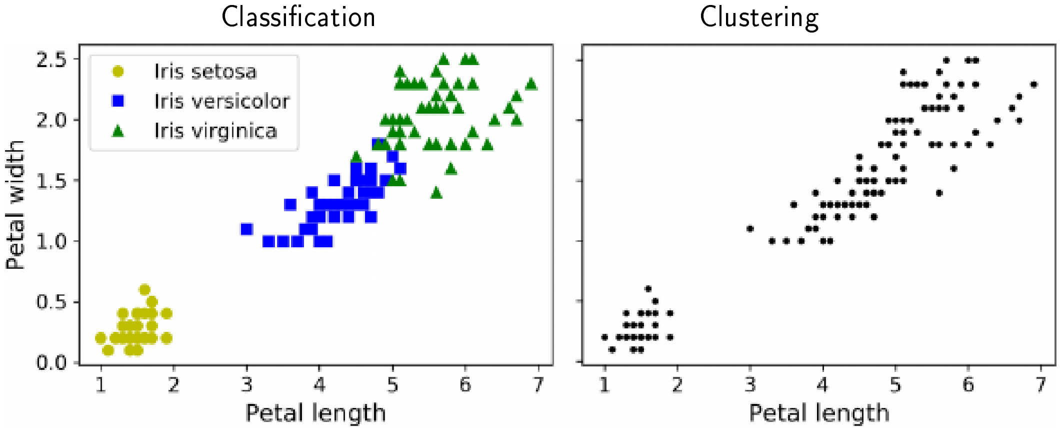
Sumary

What is Cluster Analysis?

- A **Cluster** is a collection of data objects that are:
  - Similar (or related) to each other within the same group (i.e., the cluster)
  - Dissimilar (or unrelated) to the objects in other groups (i.e., other clusters)
- **Cluster Analysis** (or **data segmentation**)
  Given a set of data points, partition them into a set of groups (clusters) that are as similar as possible.
- Cluster analysis is an **unsupervised learning** technique (there are no predefined classes).
- Typical ways to apply/use cluster analysis are:
  - As a tool to gain insights into the distribution of data
  - As part of the preprocessing step before using other algorithms.

What is Cluster Analysis

Classification vs Clustering



Cluster Analysis

Applications

- Intermediate step for other **data analysis** tasks
  - Generating a compact summary of data for classification, pattern discovery, etc.
  - Detection of *outliers*. Points far away from any cluster.
- Customer segmentation
  patients with similar symptoms or other attributes.
- Feature engineering
- Image segmentation and clustering.
- Semi-supervised learning

Cluster Analysis

Typical Clustering Methodologies

- **Distance-based Methods**:
  - Partitioning algorithms: K-means, K-medians, K-medoids
  - Hierarchical algorithms: Agglomerative vs Divisive methods
- **Density-based Methods**:
  - The data space is explored at a high level of granularity, and then post-processed to group dense regions with arbitrary shapes.
- **Probabilistic and Generative Models**: Modeling the data as a generative process
- **Clustering with High-dimensionality**: Clustering in subspaces, etc.

What is a Good Clustering?

- A good clustering method should produce "high-quality" clusters
  - **High intra-class similarity**
  - **Low inter-class similarity**
- **Quality Function**
  - It is a function that measures the "goodness of the clusters."
  - It is difficult to define "sufficiently similar" or "sufficiently good."
  - The answer is often very subjective.
- There are many similarity measures for different applications.

Sumary

Similarity Measures

Similarity, Dissimilarity, and Proximity

- **Similarity**
  - A real function that quantifies the similarity between two objects.
  - Often in the range (0,1): 0 means no similarity, 1 means identical.
- **Dissimilarity (Distance)**
  - A real function that quantifies how different two objects are.
  - Often in the range (0,1) or $(0, \infty)$: 0 means identical.
- **Proximity**: Usually refers to both similarity and dissimilarity.

Similarity Measures

Data Matrix

- Mathematically, a data point is a vector (point) in a multidimensional space, $x_i = \{x_i^1, \ldots, x_i^l\}$ where l is the dimension of the data.
- Therefore, a **dataset** of n points is a matrix of $N \times l$

$$\begin{bmatrix} x_1^1 & x_1^2 & \ldots & x_1^l \\ x_2^1 & x_2^2 & \ldots & x_2^l \\ \vdots & \vdots & \ddots & \vdots \\ x_n^1 & x_n^2 & \ldots & x_n^l \end{bmatrix}$$

Similarity Measures

Dissimilarity Matrix (Distance)

- Represents distances between two points in the dataset.
- Typically symmetric and therefore triangular.
- The **distance functions** are usually different for real, binary, categorical variables, etc.

$$\begin{bmatrix} 0 & & & \\ d(2,1) & 0 & & \\ \vdots & \vdots & \ddots & \\ d(n,1) & d(n,2) & \dots & 0 \end{bmatrix}$$

Similarity Measures

EXAMPLE: Data and Distance Matrix

- **Data Matrix**

Point	Attribute1	Attribute2
x1	1	2
x2	3	5
x3	2	0
x4	4	5

- **Distance Matrix** (Euclidean)

	x1	x2	x3	x4
x1	0			
x2	3.61	0		
x3	2.24	5.1	0	
x4	4.24	1	5.39	0

Similarity Measures

Distance in Numerical Data: Minkowski Distance

- It is one of the most popular distances:

$$d(i,j) = \sqrt[p]{|x_i^1 - x_j^1|^p + |x_i^2 - x_j^2|^p + \ldots + |x_i^l - x_j^l|^p}$$

  where $i = (x_i^1, x_i^2, \ldots, x_i^l)$ and $j = (x_j^1, x_j^2, \ldots, x_j^l)$ are two data objects of dimension l, and p is the order.
- Properties:
  - $d(i,j) > 0$ if $i \neq j$ and $d(i,i) = 0$ (Positivity)
  - $d(i,j) = d(j,i)$ (Symmetry)
  - $d(i,j) \leq d(i,k) + d(k,j)$ (Triangle Inequality)
- A distance that satisfies these properties is called a *metric*.
- $p = 2$ gives the **Euclidean distance**.

Similarity Measures

Distance in Binary Attributes

- It can be summarized in a contingency table:

	1	0
1	q	r
0	s	t

- Distance measure for symmetric binary variables:

$$d(x_i, x_j) = \frac{r+s}{q+r+s+t}$$

- Distance measure for asymmetric binary variables:

$$d(x_i, x_j) = \frac{r+s}{q+r+s}$$

Similarity Measures

Example: Distance in Binary Attributes

Name	Gender	Fever	Cough	Test1	Test2	Test3	Test4
Jaime	M	S	N	P	N	N	N
Maria	F	S	N	P	N	P	N
Juan	M	S	S	N	N	N	N

- Gender is a symmetric attribute (we do not count it)
- The rest are asymmetric
- Let S and P be "1" and N be "0"
  - $d(Jaime, Maria) = \frac{1}{1+1+1} = 0.33$
  - $d(Jaime, Juan) = \frac{1+1}{1+1+1} = 0.66$
  - $d(Juan, Maria) = ???$

Similarity Measures

Distance in Categorical Attributes

- Categorical attributes can take 2 or more states (e.g., color (red, blue, green, black))
- They are a generalization of binary attributes
- **Method 1**. Number of matches

$$d(x_i, x_j) = \frac{p - m}{p}$$

  where m is the number of matches and p is the number of attributes
- **Method 2**. Use a large number of binary attributes. Create a binary attribute for each of the M nominal states.

Similarity Measures

Distance in Ordinal Attributes

- An ordinal variable can be continuous or discrete
- The order is important: grades (fail, pass, good, excellent)
- Use a normalized rank transformation to respect the order
  - Calculate the ranks r (where $r = 1$ to M)
  - Calculate z as a scaled interval $[0, 1]$

$$z = \frac{r - 1}{M - 1}$$

  - z is now of numeric type
- Example: fail: 0; pass: 1/3; good: 2/3; excellent: 1

$$\text{dist}(\text{fail, excellent}) = 1, \ \text{dist}(\text{good, excellent}) = 1/3$$

Sumary

Partition-Based Clustering Methods

Basic Concepts

- **Partition Method**: Discovery of groupings in the data by optimizing a specific objective function and iteratively improving the quality of the partitions.
- **k-partition Method**: Partitioning a dataset **D** of **n** objects into a set of **K** clusters.
- A typical objective function: **Sum of Squared Errors**

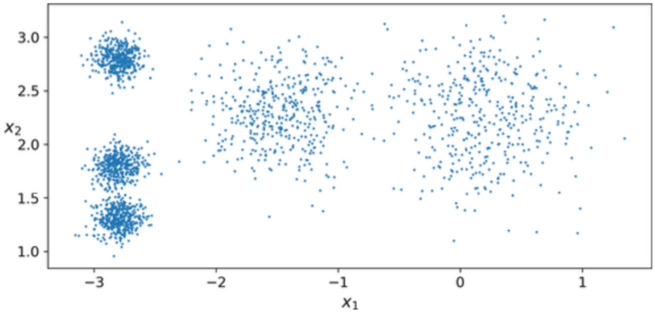
$$SSE(C) = \sum_{k=1}^{K} \sum_{x_i \in C_k} ||x_i - c_k||^2$$

where c_k is the centroid or medoid of cluster K.

Partition-Based Clustering Methods

Problem Definition: Given K, find a partition of K clusters that optimizes the chosen partitioning criterion.

- Global optimization: Requires exhaustively enumerating all partitions (not feasible)
- Heuristic methods: K-means, K-medians, K-medoids, ...



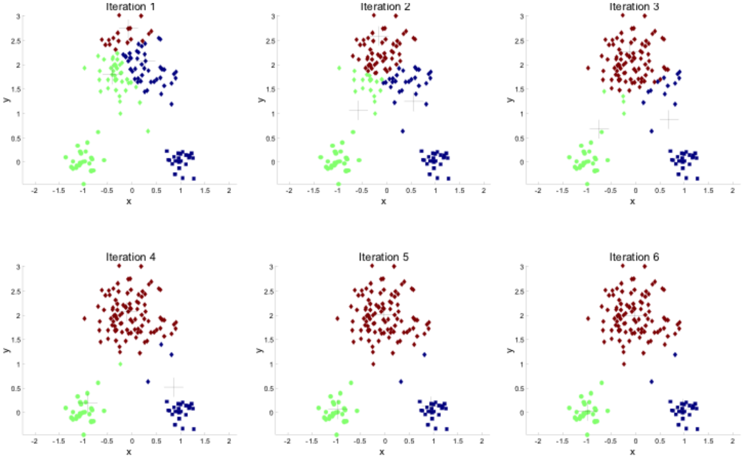
K-Means Method

Algorithm

- Each cluster is represented by the center of the cluster (centroid).
- Given K (the number of clusters), the *K-Means* algorithm works as follows:
  - Select K points as the initial centroids.
  - **Repeat**
    - Form K clusters by assigning each point to its nearest centroid.
    - Recalculate the centroids (i.e., the midpoint) of each cluster.
  - **Until** the convergence criterion is satisfied.
- Different types of similarity measures can be used (Euclidean, Manhattan, ...).

K-Means Method

Example: Execution of the K-Means Clustering Algorithm



K-Means Method

Considerations I

- **Efficiency**: $O(tkn)$ where n is the number of objects, K is the number of clusters, and t is the number of iterations.
  Normally, $K, t << n$; $\rightarrow$ it is an efficient method.
- K-Means often ends up in a **local optimum.**
  $\rightarrow$ Initialization is very important for obtaining good clusters.
- **Requires specifying K**, the number of clusters, in advance.
  $\rightarrow$ There are ways to determine the "best" K. In practice, what is done is to run the algorithm for a range of K values and select the "best" value of K.

K-Means Method

Considerations II

- **Sensitive to noise in the data and outliers.**
  $\rightarrow$ Variations: K-medoids, K-medians, ...
- K-Means can only be applied to objects with continuous numerical attributes.
  $\rightarrow$ K-Modes can be used for categorical data.
- Not suitable for discovering clusters with **non-convex shapes.**
  $\rightarrow$ Use density-based clustering.

K-Means Method

Problems I

PROBLEM 1. Sensitive to the initial choice of centroids.

POSSIBLE SOLUTION

- Perform multiple executions with various initial centroid sets and compare results.
- Estimate good centroids a priori: e.g., choose a sample and apply a hierarchical method. There are also specific methods like k-means++.

K-Means Method

Problems II

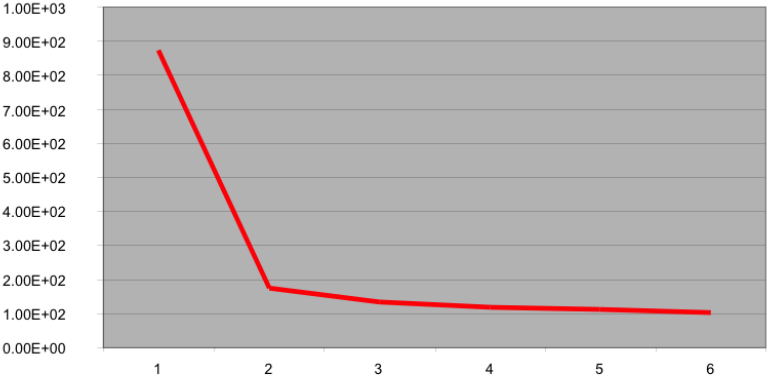
PROBLEM 2. The value of k must be chosen a priori.

POSSIBLE SOLUTION

- **Elbow Rule**: Whenever the value of k is increased, the SSE value will decrease. The normal approach is to test several values of k and check when there is no significant improvement in SSE.

K-Means Method

Problems II cont.



The elbow at k=2 suggests that this is the most appropriate value for the number of clusters.

K-Means Method

Problems III

PROBLEM 3. When using the mean to calculate centroids, the method is sensitive to outliers (anomalous values).

POSSIBLE SOLUTION

- Use medians instead of means (even with Euclidean distance).
- Preemptively remove outliers. CAUTION: Outliers can be interesting values.
- Use k-medoids: Instead of using the mean vector as a centroid, use the vector corresponding to a real data point (a representative).

K-Means Method

Problems IV

PROBLEM 4. K-Means does not work well when clusters are:

- of different sizes
- of different densities
- non-convex

POSSIBLE SOLUTION: Ad-hoc methods

Método K-Means

Recomendaciones

- **Pre-procesamiento**
  - Normalizar datos
  - Detectar outliers (eliminarlos, en su caso)
- **Post-procesamiento**
  - Eliminar pequeños clusters que puedan representar outliers.
  - Dividir clusters dispersos ('loose' clusters); esto es, clusters con un SSE relativamente alto.
  - Combinar clusters cercanos que tengan un SSE relativamente bajo.

Sumary

Cluster Analysis

Similarity Measures

Partition-Based Clustering Methods

Hierarchical Clustering: Agglomerative

Clustering Validation Methods

Hierarchical Clustering

Basic Algorithm (Agglomerative)

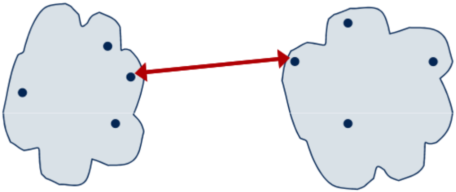
- Compute the similarity/distance matrix
- Initialization: Each data point is its own cluster
- Repeat
  - Merge the two closest clusters
  - Update the similarity/distance matrix
- Until only one cluster remains

Greedy control strategy: Once two clusters are merged, no other possible merge is reconsidered.

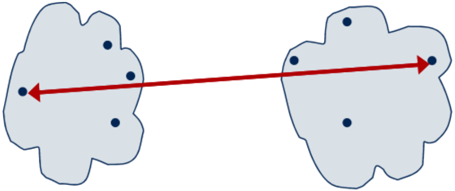
Hierarchical Clustering

How to Measure Distance Between Clusters? I

- **Single Linkage**: Distance between the closest elements of the two clusters



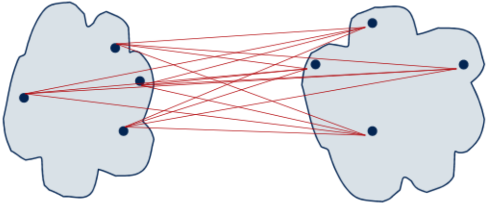
- **Complete Linkage**: Distance between the farthest elements of the two clusters



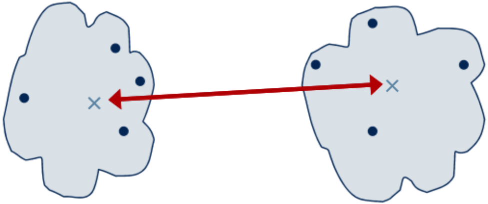
Hierarchical Clustering

How to Measure Distance Between Clusters? II

- Average of the distances between the elements of the two clusters



- Distance between the centroids of the clusters



Hierarchical Clustering

Main Drawback

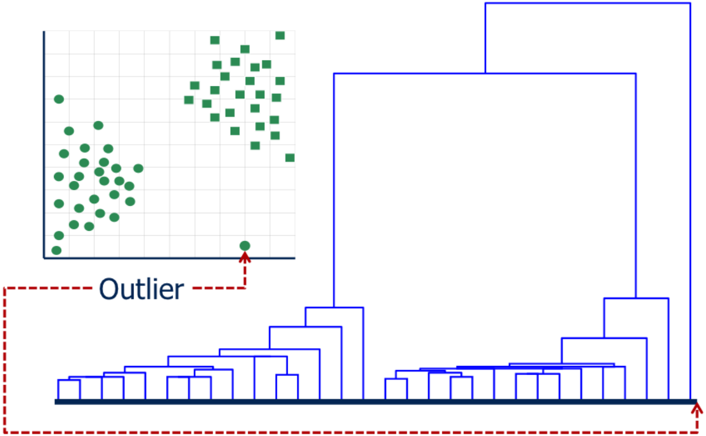
Main Drawback of Hierarchical Clustering

Low scalability: $\textit{Complexity} \geq O(n^2)$

For this reason, if a hierarchical method is used to estimate the number of clusters (for k-means), a sample of the data is typically used instead of the entire dataset.

Hierarchical Clustering

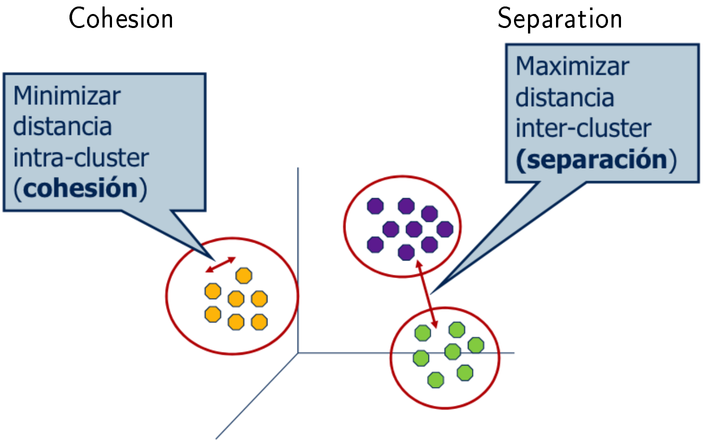
Detection of outliers.



Sumary

Clustering Validation Methods

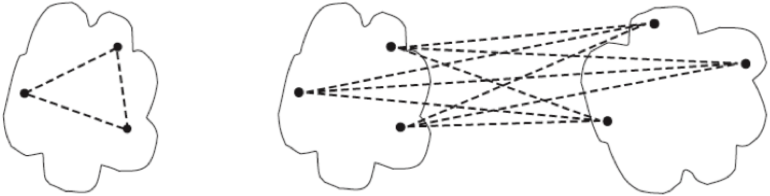
Objective of a Clustering Method



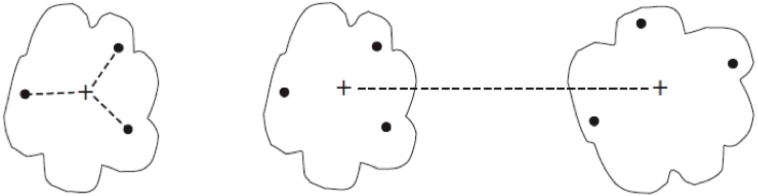
Clustering Validation Methods

Ways to Measure Cohesion and Separation

- Without using centroids



- Using centroids



Clustering Validation Methods

Euclidean Distance

- When using Euclidean distance, the use of centroids does not affect cohesion measurement.
- Minimizing cohesion and maximizing separation are equivalent.
- Therefore, SSE is a good measure of the fit (cohesion and separation) of the found centroids.

$$SSE(C) = \sum_{k=1}^{K} \sum_{x_i \in C_k} ||x_i - c_k||^2$$

Clustering Validation Methods

Quality Measures

Three categories of measures

- **External**: Supervised. Uses criteria not inherent to the dataset.
  $\Rightarrow$ Compare the clustering with prior knowledge (the ground truth).
- **Internal**: Unsupervised. Criteria derived from the data itself.
  $\Rightarrow$ Consider cohesion and separation criteria (e.g., SSE). More generally, "silhouette coefficient."
- **Relative**: For the same algorithm, directly compare different clusterings obtained with different parameters.

Clustering Validation Methods

Silhouette Coefficient

Measures the cohesion of clusters and the separation between them.

- For each point X_i, its Silhouette coefficient is

$$s_i = \frac{\mu_{out}^{in}(x_i) - \mu_{in}(x_i)}{\max\{\mu_{out}^{in}(x_i), \mu_{in}(x_i)\}}$$

  where $\mu_{in}(x_i)$ is the average distance from x_i to the points in its own cluster. $\mu_{out}^{in}(x_i)$ is the average distance from x_i to the points in the nearest cluster.
- The Silhouette coefficient (SC) is the average of the s_i values:

$$SC = \frac{1}{n}\sum_{i=1}^{n} s_i$$

- SC near $+1$ indicates good clustering (points are close to their own cluster and far from others).

Summary

Cluster Analysis

Similarity Measures

Partition-Based Clustering Methods

Hierarchical Clustering: Agglomerative

Clustering Validation Methods

Clustering

Thank you